

Solution Code - Module 3-Core Java - DecompilerDemo

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q38_DecompilerDemo.java

```
1 | import java.io.BufferedReader;
2 | import java.io.File;
3 | import java.io.FileWriter;
4 | import java.io.InputStreamReader;
5 | import java.nio.file.Files;
6 | import java.nio.file.Path;
7 | import java.nio.file.Paths;
8 |
9 | public class Q38_DecompilerDemo {
10 |     public static void main(String[] args) {
11 |         System.out.println("=== Java Decompiler Demo (using CFR) ===");
12 |
13 |         String sourceCode =
14 |             "public class DecompileTarget {\n" +
15 |             "    public void testMethod() {\n" +
16 |             "        int value = 42;\n" +
17 |             "        if (value > 10) {\n" +
18 |             "            System.out.println(\"Value is greater than 10\");\n" +
19 |             "        } else {\n" +
20 |             "            System.out.println(\"Value is small\");\n" +
21 |             "        }\n" +
22 |             "    }\n" +
23 |             "}\n";
24 |
25 |         String targetClassName = "DecompileTarget";
26 |         String srcFileName = targetClassName + ".java";
27 |         String classFileName = targetClassName + ".class";
28 |
29 |         // Find CFR decompiler jar
30 |         File cfrJar = new File("libs/cfr-0.152.jar");
31 |         if (!cfrJar.exists()) {
32 |             cfrJar = new File("week1/libs/cfr-0.152.jar");
33 |         }
34 |         if (!cfrJar.exists()) {
35 |             cfrJar = new File("../libs/cfr-0.152.jar");
36 |         }
37 |
38 |         if (!cfrJar.exists()) {
39 |             System.err.println("Error: CFR decompiler jar not found at: " + cfrJar.getAbsolutePath());
40 |             System.err.println("Please run environment setup to download it.");
41 |             return;
42 |         }
43 |     }
```

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```
44 |         try {
45 |             // 1. Write the source file
46 |             try (FileWriter writer = new FileWriter(srcFileName)) {
47 |                 writer.write(sourceCode);
48 |             }
49 |             System.out.println("1. Generated temporary class source file: " + srcFileName);
50 |
51 |             String sdkBinDir = findJavaBinDirectory();
52 |             String javacPath = sdkBinDir.isEmpty() ? "javac" : sdkBinDir + "javac";
53 |             String javaPath = sdkBinDir.isEmpty() ? "java" : sdkBinDir + "java";
54 |
55 |             // 2. Compile the class
56 |             System.out.println("2. Compiling class: " + srcFileName + "...");
57 |             runCommand(javacPath, srcFileName);
58 |
59 |             if (!new File(classFileName).exists()) {
60 |                 throw new RuntimeException("Compilation failed: class file not created.");
61 |             }
62 |             System.out.println("    Compilation succeeded.");
63 |
64 |             // 3. Decompile using CFR
65 |             System.out.println("3. Decompiling " + classFileName + " using CFR (" + cfrJar.getName() +
66 |             |
67 |             System.out.println("-----");
68 |             runCommand(javaPath, "-jar", cfrJar.getAbsolutePath(), classFileName);
69 |             System.out.println("-----");
70 |         } catch (Exception e) {
71 |             System.err.println("Decompilation demo failed: " + e.getMessage());
72 |             e.printStackTrace();
73 |         } finally {
74 |             // Cleanup files
75 |             try {
76 |                 Files.deleteIfExists(Paths.get(srcFileName));
77 |                 Files.deleteIfExists(Paths.get(classFileName));
78 |                 System.out.println("\n4. Cleaned up temporary source and class files.");
79 |             } catch (Exception ignored) {}
80 |         }
81 |     }
82 |
83 |     private static String findJavaBinDirectory() {
84 |         String path = "C:\\\\Program Files\\\\Eclipse Adoptium\\\\jdk-21.0.11.10-hotspot\\\\bin\\";
85 |         if (new File(path + "javac.exe").exists()) {
86 |             return path;
87 |         }
88 |         return "";
89 |     }
90 | }
```

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```
91 |     private static void runCommand(String... command) throws Exception {
92 |         ProcessBuilder pb = new ProcessBuilder(command);
93 |         pb.redirectErrorStream(true);
94 |         Process p = pb.start();
95 |         try (BufferedReader reader = new BufferedReader(new InputStreamReader(p.getInputStream()))) {
96 |             String line;
97 |             while ((line = reader.readLine()) != null) {
98 |                 System.out.println("    " + line);
99 |             }
100 |        }
101 |        int exitCode = p.waitFor();
102 |        if (exitCode != 0) {
103 |            throw new RuntimeException("Command failed with exit code: " + exitCode);
104 |        }
105 |    }
106 | }
```